EEE5931 Electronic Warfare Systems

Problems 2 – Radar Overview: Solutions

November 2026 · Problems from Skolnik, Introduction to Radar Systems, 3rd ed. (Ch. 1–2) · Formulas from the course formula sheet (25-5931 Formul Kagidi)

This document solves the 13 problems listed in Prroblems2-RadarOverview.pdf. The problem texts come from radar_problems.pdf. Problem 2.5 is missing from that file, so its text was taken from the textbook (Skolnik, p. 98). The formulas follow the course formula sheet and Skolnik's notation. A short list of issues found in the sources is at the end.

Constants: $c = 3\times10^8$ m/s, 1 nmi = 1852 m, 1 ft = 0.3048 m, $k = 1.38\times10^{-23}$ J/K, $(4\pi)^2 = 157.9$.

Problem 1.1 – PRF, round-trip time, range resolution, average power

> *(a) PRF for R_unamb = 60 nmi. (b) Round-trip time at R_unamb. (c) Pulse extent in space for τ = 1.5 μs. (d) Range resolution. (e) Average power for Pt = 800 kW. (f) Duty cycle.*

$R_{unamb} = 60 \times 1852 = 111\,120$ m

(a) $f_p = c / (2R_{unamb}) = 3\times10^8 / 222\,240 \approx 1350$ Hz

(b) $T_R = 2R/c = 222\,240 / 3\times10^8 = 741$ μs $\quad(= 1/f_p$, the full PRI)

(c) Extent in space $= c\cdot\tau = 3\times10^8 \times 1.5\times10^{-6} = 450$ m

(d) $\Delta R = c\cdot\tau/2 = 225$ m

(e) $P_{av} = P_t\cdot\tau\cdot f_p = 800\times10^3 \times 1.5\times10^{-6} \times 1350 \approx 1620$ W

(f) Duty cycle $= \tau\cdot f_p = 1.5\times10^{-6} \times 1350 \approx 2.0\times10^{-3} = 0.2$ %

The pulse occupies 450 m in space, but two targets only need to be cτ/2 apart: the echo from the farther target travels the extra distance twice, so the echoes separate at half the pulse length.

Answer: $f_p \approx 1350$ Hz; 741 μs; 450 m; 225 m; $P_{av} \approx 1.62$ kW; duty cycle ≈ 0.2 %

Problem 1.2 – L-band air-surveillance radar

> *f = 1300 MHz, R_max = 200 nmi for σ = 1 m², antenna 12 m × 4 m, ρ_a = 0.65, S_min = 10^{-13} W, τ = 2 μs.*

$\lambda = c/f = 3\times10^8 / 1.3\times10^9 = 0.231$ m; $\quad R_{max} = 200 \times 1852 = 370\,400$ m

(a) $A = 12 \times 4 = 48$ m²; $\quad A_e = \rho_a\cdot A = 0.65 \times 48 = 31.2$ m²

$G = 4\pi A_e/\lambda^2 = 4\pi \times 31.2 / 0.0533 \approx 7360 \;\rightarrow\; G \approx 38.7$ dB

(b) From the simple radar equation $R^4_max = P_t\cdot G\cdot A_e\cdot\sigma / ((4\pi)^2\cdot S_{min})$:

$P_t = R^4\cdot(4\pi)^2\cdot S_{min} / (G\cdot A_e\cdot\sigma) = (1.882\times10^{22})(157.9)(10^{-13}) / (7360 \times 31.2 \times 1) \approx 1.29\times10^6$ W

(c) $f_p = c/(2R_{unamb}) = 3\times10^8 / 740\,800 \approx 405$ Hz

(d) $P_{av} = P_t\cdot\tau\cdot f_p = 1.29\times10^6 \times 2\times10^{-6} \times 405 \approx 1050$ W

(e) Duty cycle $= \tau\cdot f_p = 2\times10^{-6} \times 405 \approx 8.1\times10^{-4}$ (0.081 %)

(f) $\theta_B \approx 1.22\cdot\lambda/D = 1.22 \times 0.231 / 12 = 0.0235$ rad $\approx 1.34°$

The horizontal beamwidth depends on the horizontal dimension (12 m). Skolnik's Ch. 1 rule of thumb $\theta_B \approx 65\lambda/D$ (degrees) gives 1.25°. Either value is acceptable if you state the formula used.

Answer: $A_e = 31.2$ m², $G \approx 7360$ (38.7 dB); $P_t \approx 1.29$ MW; $f_p \approx 405$ Hz; $P_{av} \approx 1.05$ kW; duty ≈ 8.1×10^{-4}; $\theta_B \approx 1.3°$

Problem 1.3 – Peak power, range and echo power vs range

> *(a) Peak power for P_av = 200 W, τ = 1 μs, f_p = 1000 Hz. (b) Range for σ = 2 m², f = 2.9 GHz, antenna 5 m × 2.7 m, ρ_a = 0.6, S_min = 10^{-13} W. (c) Sketch echo power from 10 to 80 nmi.*

(a) $P_t = P_{av} / (\tau\cdot f_p) = 200 / (10^{-6} \times 1000) = 200$ kW

(b) $\lambda = 3\times10^8 / 2.9\times10^9 = 0.1034$ m; $\quad A_e = 0.6 \times (5 \times 2.7) = 8.1$ m²

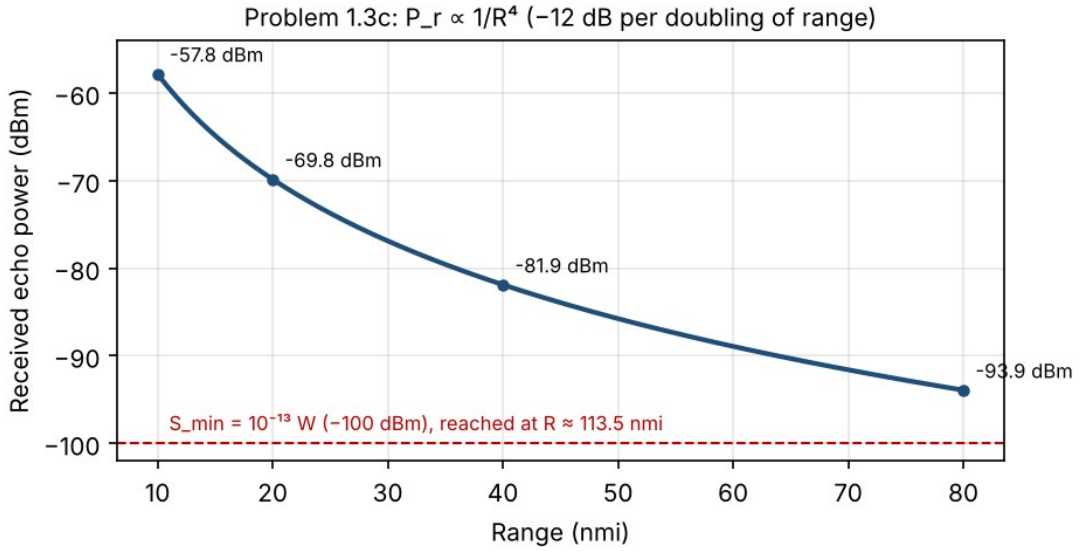
$G = 4\pi \times 8.1 / 0.1034^2 \approx 9510$ (39.8 dB)

$R^4 = P_t G A_e \sigma / ((4\pi)^2 S_{min}) = (2\times10^5)(9510)(8.1)(2) / (157.9 \times 10^{-13}) \approx 1.95\times10^{21}$ m^4

$R \approx 2.10\times10^5$ m ≈ 113.5 nmi

(c) $P_r = P_t G A_e \sigma / ((4\pi)^2 R^4)$ falls as $1/R^4$, so it drops 12 dB every time the range doubles (40 dB per decade).

R (nmi)	10	20	40	60	80
P_r (W)	1.66×10^{-9}	1.04×10^{-10}	6.5×10^{-12}	1.28×10^{-12}	4.0×10^{-13}
P_r (dBm)	−57.8	−69.8	−81.9	−88.9	−93.9



Answer: P_t = 200 kW; $R_{max} \approx 113.5$ nmi; $P_r \propto 1/R^4$ (from about −58 dBm at 10 nmi to −94 dBm at 80 nmi)

Problem 1.5 – Automotive X-band radar

> *f = 9375 MHz, τ = 10 ns, R_max = 500 ft. (a) PRF. (b) Range resolution. (c) Cross-range resolution for a 6° beam. (d) Gain of a 1 ft × 1 ft antenna, ρ = 0.6. (e) Average power for σ = 10 m², S_min = 5×10⁻¹³ W.*

$R = 500$ ft $= 152.4$ m; $\lambda = 3\times10^8 / 9.375\times10^9 = 0.032$ m

(a) $f_p = c/(2R) = 3\times10^8 / 304.8 \approx 984$ kHz

(b) $\Delta R = c\tau/2 = 3\times10^8 \times 10^{-8} / 2 = 1.5$ m

(c) $\Delta x = R\theta_B = 152.4 \times (6^\circ \times \pi/180) = 152.4 \times 0.1047 \approx 16$ m

This is **not sufficient**. A traffic lane is about 3.5 m wide, so at 500 ft the beam covers about four lanes plus the roadside. The radar cannot tell whether the echo comes from the car in its own lane, a car in the next lane, or a sign or guardrail.

(d) $A = 0.3048^2 = 0.0929$ m^2; $A_e = 0.6 \times 0.0929 = 0.0557$ m^2

$G = 4\pi A_e/\lambda^2 = 4\pi \times 0.0557 / 0.032^2 \approx 684 \rightarrow G \approx 28.4$ dB

(e) $P_t = R^4(4\pi)^2 S_{min} / (G A_e \sigma) = (152.4)^4(157.9)(5\times10^{-13}) / (684 \times 0.0557 \times 10) \approx 1.1\times10^{-4}$ W

$P_{av} = P_t \tau f_p = 1.1\times10^{-4} \times 10^{-8} \times 9.84\times10^5 \approx 1.1\times10^{-6}$ W (≈ 1.1 µW)

The required power is tiny because the range is very short and R appears to the fourth power.

Answer: $f_p \approx 984$ kHz; ΔR = 1.5 m; cross-range ≈ 16 m (not sufficient); G ≈ 28.4 dB; $P_{av} \approx 1.1$ µW ($P_t \approx 0.11$ mW)

Problem 1.6 – Minimum-cost radar design

> *f = 1230 MHz, ρ_a = 0.6, S_min = 3×10⁻¹³ W, transmitter $2.20 per watt of peak power, antenna $1400 per m², other items $1 000 000. Detect σ = 2 m² at 200 nmi.*

Step 1: relation between P_t and A. Use the simple radar equation with $G = 4\pi A_e/\lambda^2$ and $A_e = \rho_a \cdot A$:

$$R^4 = P_t \cdot G \cdot A_e \cdot \sigma \,/\, ((4\pi)^2 S_{min}) = P_t \cdot A_e^2 \cdot \sigma \,/\, (4\pi \cdot \lambda^2 \cdot S_{min})$$

$$P_t \cdot A^2 = 4\pi \cdot \lambda^2 \cdot S_{min} \cdot R^4 \,/\, (\sigma \cdot \rho_a^2) \equiv K$$

$$\lambda = 0.244\ \text{m},\ R = 370\,400\ \text{m} \ \rightarrow\ K = 4\pi(0.0595)(3\times10^{-13})(1.882\times10^{22}) \,/\, (2 \times 0.36) \approx 5.86\times10^{9}\ \text{W}\cdot\text{m}^4$$

Step 2: minimize the cost. With $P_t = K/A^2$:

$$C(A) = 2.2 \cdot K/A^2 + 1400 \cdot A + 10^6$$

$$dC/dA = -2(2.2)K/A^3 + 1400 = 0 \ \rightarrow\ A^3 = 4.4K/1400 \ \rightarrow\ A \approx 264\ \text{m}^2$$

$$P_t = K/A^2 = 5.86\times10^9 \,/\, 264^2 \approx 84\ \text{kW}$$

(a) Antenna cost = $1400 \times 264 \approx \$370\,000$; Transmitter cost = $2.2 \times 84\,000 \approx \$185\,000$

Total $\approx \$370\,000 + \$185\,000 + \$1\,000\,000 \approx \$1.55$ M

(b) At the optimum, dC/dA = 0 gives $1400 \cdot A = 2 \times (2.2 \cdot P_t)$: **antenna cost = 2 × transmitter cost**. As a first attempt, put about two thirds of the antenna-plus-transmitter budget into the antenna and one third into the transmitter. This works because $P_t \propto 1/A^2$, so money spent on the antenna reduces the required power quadratically. (Check: $G = 4\pi\rho_a A/\lambda^2 \approx 45$ dB.)

Answer: A ≈ 264 m², P_t ≈ 84 kW; antenna ≈ $370 k, transmitter ≈ $185 k; rule: antenna cost ≈ 2 × transmitter cost

Problem 1.9 – Minimum-weight radar

> *W_T = k_T·P, W_A = k_A·A^(3/2). Find the relation between W_A and W_T that minimizes W = W_T + W_A at fixed range.*

As in Problem 1.6, at a fixed range the radar equation gives $P \cdot A^2 = K$ (constant), so $P = K/A^2$.

$$W(A) = k_T \cdot K/A^2 + k_A \cdot A^{(3/2)}$$

$$dW/dA = -2k_T \cdot K/A^3 + (3/2)k_A \cdot A^{(1/2)} = 0$$

Multiply by A: $-2(k_T \cdot K/A^2) + (3/2)(k_A \cdot A^{(3/2)}) = 0$, that is $-2W_T + (3/2)W_A = 0$:

$$W_A = (4/3) \cdot W_T$$

In general, if the antenna cost or weight scales as A^m and $P \propto 1/A^2$, the optimum is (antenna share) = (2/m) × (transmitter share). With m = 1 this gives the factor 2 of Problem 1.6; with m = 3/2 it gives 4/3.

Answer: Minimum total weight when $W_A = (4/3) \cdot W_T$: the antenna weighs one third more than the transmitter.

Problem 2.5 – False-alarm time and threshold

> *T_fa = 30 min, B = 0.4 MHz. (a) P_fa. (b) Threshold-to-noise ratio V_T^2/ψ_0. (c) Repeat for T_fa = 1 year (8760 h). (d) Threshold set 0.3 dB lower than (b): new T_fa? (e) Threshold 0.3 dB higher: new T_fa? (f) Comment on achieving a specified T_fa precisely. [Text from Skolnik p. 98]*

Formula sheet: $P_{fa} = 1/(T_{fa} \cdot B)$ and $T_{fa} = (1/B) \cdot \exp(V_T^2/2\psi_0)$, so $V_T^2/\psi_0 = 2 \cdot \ln(T_{fa} \cdot B)$.

(a) $T_{fa} \cdot B = 1800 \times 4\times10^5 = 7.2\times10^8 \ \rightarrow\ P_{fa} = 1/(7.2\times10^8) \approx 1.39\times10^{-9}$

(b) $V_T^2/\psi_0 = 2 \cdot \ln(7.2\times10^8) = 2 \times 20.40 = 40.8 \ \rightarrow\ 16.1$ dB

(c) $T_{fa} = 8760 \times 3600 = 3.15\times10^7$ s; $T_{fa} \cdot B = 1.26\times10^{13} \ \rightarrow\ P_{fa} \approx 7.9\times10^{-14}$

$V_T^2/\psi_0 = 2 \cdot \ln(1.26\times10^{13}) = 60.3 \ \rightarrow\ 17.8$ dB

(d) −0.3 dB: $V_T^2/\psi_0 = 40.8 \times 10^{(-0.03)} = 38.07 \ \rightarrow\ T_{fa} = \exp(19.03)/(4\times10^5) \approx 461\ \text{s} \approx 7.7\ \text{min}$

(e) +0.3 dB: $V_T^2/\psi_0 = 40.8 \times 10^{(0.03)} = 43.71 \ \rightarrow\ T_{fa} = \exp(21.85)/(4\times10^5) \approx 7740\ \text{s} \approx 2.15\ \text{h}$

(f) The false-alarm time depends exponentially on the threshold. A change of only ±0.3 dB moves T_{fa} from 30 min to between 8 min and 2 h. Going from 30 minutes to one year (17 500 times longer) needs only 1.7 dB more threshold. Setting a specific T_{fa} precisely would require knowing the noise level and receiver gain to a small fraction of a dB, which is not practical with drift, temperature changes and interference. Real radars therefore use CFAR (constant false-alarm rate) processing, which estimates the noise level and adjusts the threshold automatically.

Answer: $P_{fa} \approx 1.4\times10^{-9}$, $V_T^2/\psi_0 \approx 40.8$ (16.1 dB); 1 year: $P_{fa} \approx 7.9\times10^{-14}$, 60.3 (17.8 dB); −0.3 dB → ≈ 7.7 min; +0.3 dB → ≈ 2.2 h; T_{fa} is extremely sensitive to the threshold.

Problem 2.8 – Pulses per scan and integration

> *f = 1.35 GHz, D = 32 ft, R_unamb = 220 nmi, scan time 10 s. (a) Number of pulses per scan (θ_B = 1.2λ/D). (b) Integration loss and integration-improvement factor for P_d = 0.9, P_fa = 10^{-6}.*

$$\lambda = 3\times10^8 / 1.35\times10^9 = 0.222 \text{ m}; \quad D = 32 \times 0.3048 = 9.75 \text{ m}$$

$$\theta_B = 1.2\lambda/D = 1.2 \times 0.222 / 9.75 = 0.0273 \text{ rad} = 1.57^\circ$$

$$f_p = c/(2R_{unamb}) = 3\times10^8 / (2 \times 407\,440) \approx 368 \text{ Hz}$$

$$\dot{\theta}_s = 360^\circ / 10 \text{ s} = 36^\circ/\text{s}$$

$$\text{(a) } n = \theta_B \cdot f_p / \dot{\theta}_s = 1.57 \times 368 / 36 \approx 16 \text{ pulses}$$

(b) From Skolnik Fig. 2.7 (square-law detector, Marcum), at n = 16, $P_d = 0.9$ and $n_f = 1/P_{fa} = 10^6$ (between the 10^4 and 10^{12} curves):

$$I_i(16) \approx 9 \ (\approx 9.5 \text{ dB}); \quad L_i(16) \approx 2.5 \text{ dB}$$

Numerical check (noncoherent integration of 16 pulses, nonfluctuating target): the single-pulse requirement is $(S/N)_1 = 13.2$ dB and the per-pulse requirement with 16 pulses is $(S/N)_{16} = 3.8$ dB. Hence:

$$I_i = (S/N)_1/(S/N)_{16} = 13.2 - 3.8 = 9.4 \text{ dB} \ (\approx 8.6); \quad E_i = I_i/n \approx 0.54; \quad L_i = 10\log(1/E_i) \approx 2.7 \text{ dB}$$

Interpretation: with perfect (coherent) integration, 16 pulses would give a 12 dB improvement. Noncoherent integration gives about 9.4 dB, so about 2.5–2.7 dB is lost.

Answer: **n ≈ 16 pulses; $I_i \approx 9$ (≈ 9.4 dB); $L_i \approx 2.5$–2.7 dB**

Problem 2.10 – Maximum RCS of a metallic sphere

In Skolnik Fig. 2.8, the RCS of a sphere peaks in the resonance (Mie) region at $2\pi a/\lambda = 1$, where the circumference equals one wavelength. The peak is 5.6 dB above the optical-region value, so $\sigma_{max} \approx 3.6\cdot\pi a^2$.

$$\text{(a) } D = 1 \text{ m } (a = 0.5 \text{ m}): \lambda = 2\pi a = \pi \text{ m} \approx 3.14 \text{ m} \rightarrow f = c/\lambda \approx 95.5 \text{ MHz} \quad (\sigma_{max} \approx 3.65 \times 0.785 \approx 2.9 \text{ m}^2)$$

$$\text{(b) } D = 1 \text{ mm } (a = 0.5 \text{ mm}): \lambda = \pi \text{ mm} \rightarrow f \approx 95.5 \text{ GHz} \quad (\sigma_{max} \approx 2.9\times10^{-6} \text{ m}^2)$$

The result scales with size: making the sphere 1000 times smaller moves the peak 1000 times higher in frequency.

Answer: **(a) ≈ 95.5 MHz; (b) ≈ 95.5 GHz**

Problem 2.11 – License plate RCS

> *Plate 12 in × 6 in, f = 10.525 GHz. (a) Maximum RCS. (b) Vertical tilt for −10 dB. (c) Other contributors to a car's frontal RCS.*

$$\lambda = 3\times10^8 / 10.525\times10^9 = 0.0285 \text{ m}; \quad A = 0.3048 \times 0.1524 = 0.0465 \text{ m}^2$$

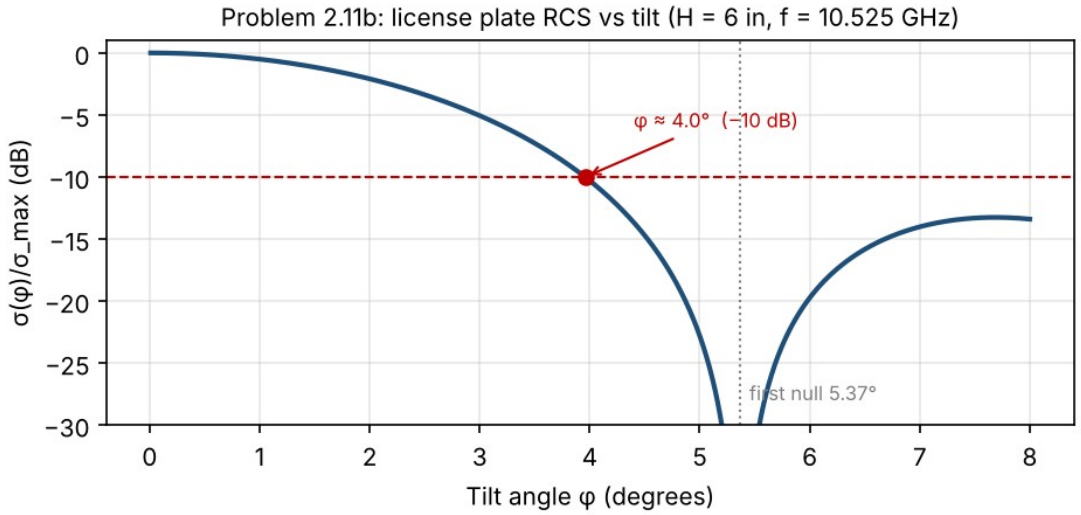
$$\text{(a) } \sigma_{max} = 4\pi A^2/\lambda^2 = 4\pi \times 0.0465^2 / 0.0285^2 \approx 33.4 \text{ m}^2 \ (15.2 \text{ dBsm})$$

A small flat plate is a strong reflector at normal incidence: 33 m² is comparable to a whole aircraft.

(b) With H = 6 in = 0.1524 m, $H/\lambda = 5.35$. Let $x = 2\pi(H/\lambda)\cdot\sin\phi$. Find x where $\sin^2 x / x^2 = 0.1$ (−10 dB):

$$\sin x / x = 0.316 \rightarrow x \approx 2.32 \text{ rad}$$

$$\sin\phi = x / (2\pi \times 5.35) = 2.32 / 33.6 = 0.069 \rightarrow \phi \approx 4.0^\circ$$



The first null is at x = π (ϕ ≈ 5.4°), so a tilt of only about 4° reduces the RCS by 10 dB.

(c) Other contributors from the front: the headlight reflectors (parabolic mirrors behind the lens act like corner and dish reflectors), the bumper and grille, the radiator and engine block seen through non-metallic parts, dihedral and corner structures (bumper–body joints, hood–windshield junction, wheel wells), the windshield frame, side mirrors, and the license plate bracket. Multiple reflections between the car and the road surface also add to the echo.

Answer: σ_{max} ≈ 33 m²; tilt ≈ 4° for −10 dB; headlights, grille/bumper, radiator/engine, corner-like structures, mirrors

Problem 2.18 – Single-pulse SNR: nonfluctuating vs Swerling 1

$P_{fa} = 10^{-6}$, detection on a single pulse. The nonfluctuating values come from Skolnik Fig. 2.6 (equivalently Albersheim's equation or the exact Marcum calculation). For a Swerling 1 target with one pulse there is an exact closed form: $P_d = P_{fa}^{\wedge}(1/(1+S/N))$, so $S/N = \ln(P_{fa})/\ln(P_d) - 1$.

(a) Nonfluctuating, $P_d = 0.50$: S/N ≈ 11.2 dB

(b) Nonfluctuating, $P_d = 0.99$: S/N ≈ 14.5 dB

(c) Swerling 1, $P_d = 0.50$: $S/N = \ln(10^{-6})/\ln(0.5) - 1 = 19.93 - 1 = 18.9$ → 12.8 dB

Swerling 1, $P_d = 0.99$: $S/N = 13.82/0.01005 - 1 \approx 1374$ → 31.4 dB

P_d	Nonfluctuating	Swerling 1	Fluctuation loss L_f
0.50	11.2 dB	12.8 dB	≈ 1.5 dB
0.99	14.5 dB	31.4 dB	≈ 17 dB

(d) Conclusions:

- For a nonfluctuating target, going from $P_d = 0.5$ to 0.99 costs only about 3 dB.
- For a fluctuating (Swerling 1) target, the same step costs about 19 dB, because a deep fade of the echo occasionally makes detection impossible no matter how good the average SNR is.
- At $P_d = 0.5$ the fluctuation loss is small (~1.5 dB), but at high P_d it is very large (~17 dB). Demanding very high single-pulse P_d against fluctuating targets is extremely expensive.
- In practice, high P_d is obtained by integrating several pulses and by decorrelating the target fluctuations (for example frequency agility or diversity), not by raising the power.

Answer: 11.2 / 14.5 dB (nonfluctuating); 12.8 / 31.4 dB (Swerling 1); fluctuation loss grows from ~1.5 dB to ~17 dB

Problem 2.19 – Resolving range ambiguity with two PRFs

$$R_{u1} = c/(2{\cdot}4000) = 37.5 \text{ km} = 20.25 \text{ nmi}; \quad R_{u2} = c/(2{\cdot}3500) = 42.86 \text{ km} = 23.14 \text{ nmi}$$

The true range must satisfy both measurements: $R = 7 + m{\cdot}20.25 = 18.6 + k{\cdot}23.14$ for integers m and k. Try m = 0, 1, 2, …

m	R = 7 + 20.25m (nmi)	R − 18.6	Multiple of 23.14?
1	27.25	8.65	No
2	47.50	28.90	No
3	67.75	49.15	No (2 × 23.14 = 46.28)
4	88.00	69.40	Yes (3 × 23.14 = 69.42)

$$R_{true} = 7 + 4 \times 20.25 = 18.6 + 3 \times 23.14 \approx 88 \text{ nmi}$$

Answer: True range ≈ 88 nmi

Problem 2.20 – Cosecant-squared antenna pattern

(a) For an aircraft at constant height h over a flat earth, the elevation angle satisfies $\sin\phi = h/R$, so $\csc\phi = R/h$. With the same antenna for transmit and receive:

$$P_r = P_t G^2 {\cdot} \lambda^2 {\cdot} \sigma \,/\, ((4\pi)^3 R^4), \quad G = G_o {\cdot} \csc^2\phi = G_o {\cdot} (R/h)^2$$

$$P_r = P_t G_o^2 {\cdot} (R/h)^4 {\cdot} \lambda^2 {\cdot} \sigma \,/\, ((4\pi)^3 R^4) = P_t G_o^2 {\cdot} \lambda^2 {\cdot} \sigma \,/\, ((4\pi)^3 h^4)$$

The R^4 terms cancel, so the received power depends only on the height h and **not on range R**.

(b) A simple fan beam that is narrow in elevation leaves a gap at high elevation angles: an aircraft that comes close to the radar at altitude leaves the beam. The $\csc^2$ pattern spreads the energy in elevation so that coverage extends up to a constant-altitude ceiling at all ranges. It also avoids wasting power at short range and high angles, where a fan beam would give much more signal than needed.

(c) Limitations of the simple result:

- The earth is not flat: curvature and atmospheric refraction change the geometry at long range.
- Ground reflections (multipath) create lobes and nulls in the elevation pattern.
- A real antenna only approximates $\csc^2$ over a limited range of angles (G cannot become infinite as $\phi \to 0$). The shaped beam also has lower peak gain (about 2 dB) than an unshaped fan beam of the same size.
- The target RCS is not constant: it changes with aspect angle and fluctuates.
- Aircraft do not fly at exactly constant height, and the effect of the antenna's azimuth pattern and propagation losses is ignored.

Answer: $P_r = P_t {\cdot} G_0^2 {\cdot} \lambda^2 {\cdot} \sigma / ((4\pi)^3 h^4)$, independent of R; $\csc^2$ gives coverage up to a constant altitude without wasting power; limitations: earth curvature, multipath, real antenna patterns, RCS variation

Issues Found in the Sources

Where	Issue	Comment
radar_problems.pdf	Problem 2.5 is missing (the Chapter 2 list starts at 2.7).	Text taken from Skolnik, 3rd ed., p. 98.
Formula sheet	The integration line reads "$E_i(n) = (S/N)_n = (S/N)_1/n$".	$(S/N)_n = (S/N)_1/n$ holds only for ideal (coherent) integration. The definition is $E_i(n) = (S/N)_1 / [n\cdot(S/N)_n]$ (Skolnik Eq. 2.32), with $I_i = n\cdot E_i$ and $L_i = 10\cdot\log(1/E_i)$.
Problem 1.2f / formula sheet	Beamwidth constant differs between sources: 1.22λ/D (formula sheet), 1.2λ/D (Problem 2.8), 65λ/D in degrees (Skolnik Ch. 1).	Results differ by ~10 %. State which formula you use.
Problem 2.8b	Skolnik Fig. 2.7 uses Marcum's false-alarm number n_f; reading it at $n_f = 10^6$ requires interpolating between the 10^4 and 10^{12} curves.	Figure gives $I_i \approx 9$, $L_i \approx 2.5$ dB; the exact calculation gives 8.6 and 2.7 dB.

Sources

- M. I. Skolnik, Introduction to Radar Systems, 3rd ed., McGraw-Hill, 2001: Ch. 1, Ch. 2 (Eqs. 2.21, 2.32; Figs. 2.6, 2.7, 2.8; problems p. 98–103).
- EEE5931 course notes 26-03 Radar Overview and the course formula sheet (25-5931 Formul Kagidi).